Storyboard: Who powers France?

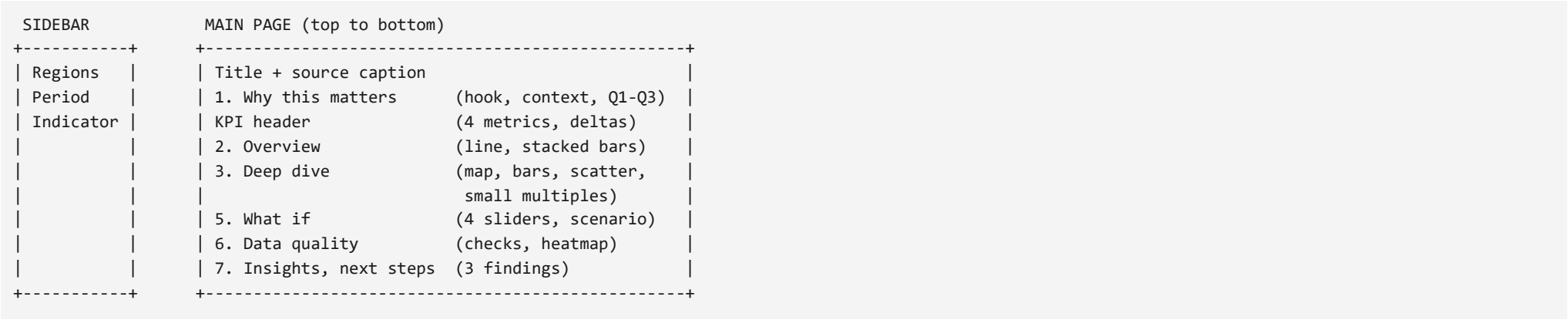
Individual project - Data Visualisation (EFREI, #EFREIDataStories2026). Streamlit dashboard on regional electricity production and consumption (RTE eCO2mix, 12 metropolitan regions, 2013 to 2026).

Story in one line

Electricity is produced and consumed locally, and French regions are very different: a few power the others, and a green mix does not mean self-sufficiency.

- **Audience**: citizens, students and regional decision makers who want a quick, visual answer.
- **Narrative pattern**: compare regions (map, ranking, small multiples) + change over time + what-if exploration.
- **Questions**:
    1. Which regions produce more electricity than they consume?
    2. How green is each regional mix, and how did it change?
    3. What would change if a region built more solar or wind?

Flow of the page



The sidebar filters drive the KPIs and sections 2, 3 and 5. Section 7 always compares the first and last complete years, whatever the filters.

Step by step

App no.	Section	Role in the story	Message (headline)	Visual	Interaction	Question answered
1	1. Why this matters	**Hook and context**	France is seen as a low-carbon exporter, but regions differ a lot: some supply the country, others import most of what they use.	Text, two columns, info box	Sidebar filters introduced	Why should I care?
-	KPI header	**Key numbers**	Where do the selected regions stand today?	4 metrics: renewable share, CO2 intensity, self-sufficiency, production	Tied to the filters; change vs the first 12 months of the selection	What is the situation now, and is it improving?
2	2. Overview	**Change over time**	The selected indicator went from X to Y; nuclear and renewables share the yearly mix.	Line chart (12-month rolling, one grey line per region, bold total); stacked bars by source per year	Hover tooltips, indicator and period from sidebar	Q2: how did it change?
3	3. Deep dive: compare regions	**Comparison**	One region ranks first, another last; N regions produce more than they consume.	Choropleth map, ranking bars with reference line, scatter (self-sufficiency vs CO2 intensity, sized by production), small multiples	Hover, radio button (TWh or %) for small multiples	Q1: who produces more than they consume? Q2: how green is each mix?
5	5. What if	**Exploration**	The same percentage increase brings very different gains depending on the region's starting point.	3 metrics with deltas, stacked bars today vs scenario, dashed consumption line	Region select, 4 sliders (solar, wind, nuclear, thermal)	Q3: what if a region built more solar or wind?
6	6. Data quality and limitations	**Trust**	The data is consistent enough for monthly comparisons; glitches are documented.	Validation table, missing-values heatmap, limitations list	Hover	Can I trust these numbers?
7	7. Key insights and next steps	**Insights and implications**	The mix got cleaner; a few regions power the others; a high renewable share does not mean self-sufficiency.	3 insight boxes, implications, next steps	None	What should I remember, and what next?

Key insights (section 7)

1. **The mix got cleaner**: renewable share up and estimated CO2 intensity of production down between the first and last complete years.
2. **A few regions power the others**: some regions have a self-sufficiency above 100 %, mainly thanks to nuclear.
3. **A high renewable share does not mean self-sufficiency**: a small renewable production gives a high share but little autonomy (Île-de-France as the extreme case).

Design choices

- One **headline sentence** per section, computed from the filtered data.
- Every chart has a **text description**, labelled axes with units, and a colour-blind-safe palette for the sources.
- Help text on filters and KPIs; every control changes at least one chart.
- Assumptions (fixed emission factors, constant consumption in the what-if, missing values counted as 0) are written next to the results they affect.

Demo video outline (2 to 4 minutes)

Time	What to show
0:00 - 0:30	Hook and questions (section 1), sidebar filters
0:30 - 1:00	KPI header, change the period to show the deltas
1:00 - 1:45	Overview: switch the indicator, read the trend and the mix by year
1:45 - 2:30	Map, ranking, scatter, small multiples (absolute vs share)
2:30 - 3:15	What-if: add solar in one region, then compare with another region
3:15 - 3:45	Data quality section and limitations
3:45 - 4:00	Insights and next steps